Lonnie VanZandt

Curriculum Vitae — April 2026

Enterprise Architect · Systems Engineer · Founder, DPYC Network State Vergennes / Panton, Vermont

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Profile

Enterprise Architect, Systems Engineer, and Software Engineer with thirty-eight years of experience in mission-critical software, enterprise architecture, and model-based systems engineering across defense, telecommunications, aerospace, and commercial programs. In 2026 founded the DPYC™ (Don't Pester Your Customer™) Network State — a decentralized API monetization economy grounded in Bitcoin Lightning, Nostr-native identity, Austrian sound-money economics, and the Model Context Protocol — and serves as its Prime Authority. Principal author of the Tollbooth DPYC™ protocol and of multiple open-source MCP servers that implement it. OMG-certified modeler, Neo4j-certified professional, and contributing author to international standards bodies. Applied cybersecurity domains include CMMC, NIST, STIGs, FIPS, the Risk Management Framework, SBOM generation, and dependency vulnerability scanning in Java and Scala CI/CD pipelines. Current research spans Collaborative Ontology Engineering, Decision Management, Provenance Management, and Argumentation and Skepticism in Collaborative Engineering Artifacts. Holds a recent U.S. security clearance.

Professional Experience

Founder & Prime Authority — DPYC Network State `FOUNDER`

2026 – Present · Panton, Vermont

- Conceived, designed, and launched DPYC™ (Don't Pester Your Customer™) — a decentralized API monetization protocol built on Bitcoin Lightning, Nostr-native identity (npubs), and the Model Context Protocol. DPYC is a Network State in the sense Balaji Srinivasan gives the term: a coordinated online community with its own governance, economy, and membership.
- Holds trademark on DPYC™, Tollbooth DPYC™, and Don't Pester Your Customer™.
- Serves as Prime Authority (First Curator) of the DPYC Social Contract governance structure; oversees Authority certification, Operator registration, and community governance via the `lonniev/dpyc-community` GitHub registry.
- Architected the Tollbooth DPYC™ protocol: pre-funded Lightning balances eliminate per-request payment negotiation; kind-30079 Nostr-signed certificates enforce metered API access; a composable Constraint Engine imposes temporal, supply, rate-limit, surge, and promotional pricing rules.
- Designed a fractal, self-similar multi-Authority hierarchy enabling arbitrarily deep trust chains without specialized processing.
- Designed the network-effect moat: all software is released under Apache 2.0, yet replication of the software in isolation is economically futile because value accrues to the network — to its Authorities, Operators, Patrons, and the shared ledger — rather than to any individual package. The design allows the architects of the economy to receive compensation through network participation even though the code remains fully open.
- Applied Austrian praxeology and sound-money economics as the philosophical foundation for honest, non-predatory, demurrage-based API credit systems.
- Authored the provisional patent cover memo and figure set for the Tollbooth DPYC Economic Architecture (six claim families, filing target mid-2026).
- Operates a BTCPay Server Lightning node on AWS EC2 as the sole payment infrastructure for the network.
- Operates the live fleet on Prefect Horizon: six monetized Operators — TheBrain, Schwab, eXcalibur, TaxSort, Optionality, and a weather-stats teaching sample — plus the free DPYC Oracle and two community-utility MCPs (an OAuth2 callback collector and a URL shortener).

Store Owner and Administrator — Good Brew

2026 – Present · stablecoin.myshopify.com

- Founded and operates Good Brew, a Shopify print-on-demand merchandise store for DPYC Network State branded goods.
- Responsible for Shopify store administration, product catalog management, Gelato Print-on-Demand integration, BTCPay Server Bitcoin and USDC payment processing, and Stripe fiat settlement.
- DPYC logo designed in-house; SVG assets maintained in the `dpyc-community` repository.

Author — Monetized MCP Servers `OPEN SOURCE`

2026 – Present · github.com/lonniev

- **thebrain-mcp** — FastMCP Python server monetizing AI-agent access to a TheBrain knowledge graph of over nine thousand thoughts via the Tollbooth DPYC protocol; hosted on Prefect Horizon.
- **schwab-mcp** — Multi-patron metered gateway to the Charles Schwab brokerage API; full OAuth2 Authorization Code flow handled in-MCP via `schwab_begin_oauth` and `schwab_check_oauth_status`; Tollbooth credit billing per call.
- **excalibur-mcp** — X / Twitter posting MCP server with Tollbooth billing and NIP-17 Secure Courier credential delivery; supports Unicode Mathematical Alphanumeric Symbol rich-text formatting.
- **taxsort-mcp** — Personal tax-transaction classification with bulk CSV import and IRS category mapping; paired with a React 18 + Vite + TypeScript frontend deployed on Cloudflare Pages.
- **optionality-mcp** — An AI-judged options-trading drill that grades practice trades across multiple dimensions; paired with a React 18 + Vite + TypeScript frontend on Cloudflare Pages.
- **tollbooth-sample** — An educational weather-stats Operator; the canonical reference implementation new Operators are scaffolded from.
- Beyond the Operators, authored two community-utility (Advocate) MCPs: **tollbooth-oauth2-collector** (a shared, unauthenticated OAuth2 callback mailbox) and **tollbooth-shortlinks** (ephemeral, Nostr-friendly short URLs).
- All Operators implement the Tollbooth Patron Credential Architecture: NeonVault-backed credential storage, BTCPay Lightning invoicing, and per-field regex credential delivery.
- Python maintainer of the **tollbooth-dpyc** wheel — the shared library underlying all Tollbooth-compliant MCP servers.

Swift and iOS Developer — Pricing Studio

2026 – Present · DPYC Ecosystem · iOS / SwiftUI

- Designing and developing Pricing Studio, a native iOS application enabling DPYC Operators to configure, visualize, and publish Tollbooth pricing models.
- Implements the client-side Constraint Engine for rate-bps floors, ceilings, surge multipliers, and per-tool pricing rules surfaced through a SwiftUI interface, with NIP-98-authenticated deployment to live MCP endpoints.

Chief Solutions Officer — Intercax LLC

January 2026 – Present · Vermont, United States (Remote)

Promoted from Principal Solution Architect in January 2026.

- Leads the Customer Support Team, the Quality Assurance Team, and the DevSecOps IT Team in delivering innovative business solutions built on Syndeia, Intercax's digital-thread platform for aerospace, defense, automotive, healthcare, and energy sectors.

Principal Solution Architect — Intercax LLC

October 2020 – January 2026 · Vermont, United States (Remote)

- Led the Customer Support Team for Intercax, assuring user success with Syndeia and fostering long-term relationships with clients.
- Led Intercax's CMMC and NIST cybersecurity initiatives to protect Controlled Unclassified Information (CUI).

- Led the Kubernetes and Docker infrastructure initiatives for Syndeia.
- Served as Technical Lead on the Business Development Team and as Technical Account Manager for Intercax's primary services contracts.
- Served as Subject Matter Expert for STIGs, FIPS, and Java Security; for Apache TinkerPop Gremlin graph query; for OAuth2 and SSO SAML2 authentication; and for the business value of Digital Threads in Digital Engineering.
- Mentored U.S. Defense Industrial Base contractors on digital engineering, digital threading, and API integrations.
- Software supply-chain and CI/CD modernization: introduced Maven for the Java SDLC; improved the Scala SBT build; implemented SBT publishing to GHCR and AWS ECR; introduced Snyk and OWASP Dependency-Check scanning; introduced CycloneDX SBOMs to SBT builds; introduced GitHub Actions for cloud-based CI/CD.
- Applied expertise in OSLC, Jazz DNG and GCM, and Windchill; data transformation using HTTP/REST, XML, XMI, JSON, OWL, RDF, and Linked Data Platform.
- Co-authored and led Intercax's first NASA Phase 2 SBIR.
- Presented at INCOSE, OMG, NDIA, and the Intercax Digital Thread Conference series on Syndeia and collaborative engineering.
- Developed in Scala, Java, and Python.

AWS Solutions Consultant — Independent

January 2025 · Remote

- Implemented an AWS Amplify site containing an embedded Amazon Bedrock AI chat bot.
- Implemented a complete CloudFormation stack for an ECS Fargate Dockerized service fronted by AWS Application Load Balancers and API Gateways.
- Implemented AWS Cognito identity provisioning with OIDC and OAuth2 proxies.
- Implemented a complete AWS Marketplace product offering for a user-metered container product, including custom Lambda functions that release CloudFormation templates from S3 only when a subscription is verified active.
- Integrated a Scala build process with AWS ECR to generate Docker images for ECS Fargate in the CI/CD pipeline.

Business Analyst T5 — CACI International Inc.

June 2018 – October 2020 · Chantilly, VA

- Lead Subject Matter Expert on Enterprise Architecture, focusing on UPDM, UAF, and SysML, for CACI's Department of Defense acquisition programs.
- Lead Analyst for several of CACI's classified C4ISR projects.
- Subject Matter Expert on the Cameo MagicDraw OpenAPI for plugin customization.
- MBSE trainer to CACI staff and U.S. Navy clients.
- Refactored the UAF Security Viewpoint back from UAF to UPDM to make Security Viewpoints available to CACI's UPDM-anchored clients.
- Risk Management Framework Subject Matter Expert.
- Senior contributor to CACI's center of excellence for model-based systems engineering, advancing digital-engineering practices and solution patterns.

Enterprise Architect — SODIUS SAS

February 2014 – June 2018 · Evergreen, CO

- Contributing software engineer for OSLC integration products across bespoke lifecycle-management engineering tools.
- Subject Matter Expert for Cameo MagicDraw; for IBM Jazz DNG and Windchill PDMLink integration via OSLC; and for RDF Linked Data Tracked Resource Sets.
- Implemented extensive customization of Windchill deployments and its Sencha user interface to expose OSLC capabilities.

- Automated the deployment of the complete Windchill PDMLink service and the IBM Jazz Engineering Lifecycle Management suite using Ruby Vagrant.
- Product Portfolio Technology Planner and proponent of Aha! for technology roadmaps; Agile ScrumMaster; Business Development Associate.

Chief Architect — No Magic, Inc.

August 2012 – February 2014 · Evergreen, CO

- Liaison between customers, users, engineering teams, marketing and sales, and executive management.
- Forward-deployed Applications Engineer for MagicDraw and UPDM.
- Established solution strategies that supported customer satisfaction across legacy, current, and future products.
- Represented No Magic as technology evangelist for enterprise architectures and for systems engineering using UPDM and SysML; served at the Object Management Group on the UML, SysML, and UPDM revision task forces.
- Led the initial implementation of Teamwork Cloud, replacing the less-scalable Teamwork Server.

Systems Engineer — LGS Innovations

December 2008 – August 2012 · Westminster, CO

- Systems Engineering Subject Matter Expert supporting the IR&D Bid and Proposal Team with technology insight, competitive analysis, customer expectations, product reliability, and requirements clarification.
- Contributed to numerous classified Technical Task Orders ranging in value from $250,000 to over $5,000,000 for innovative tactical base-station systems spanning GSM, CDMA, UMTS, HSPA, LTE, and WiMAX air interfaces.
- RF Link Budget analysis; proposal author; Requirements Engineer; Lead Technical Writer.

Consulting Applications Engineer — Artisan Software Tools Limited

March 2007 – December 2008 · North America

- Consulting Applications Engineer for North America; technical sales support and professional services for aerospace and defense systems engineers.
- Tools services for UML, SysML, MARTE, and DoDAF modeling and for Model-Driven Design and MDA targeting C, C++, Java, and Ada software systems.
- Specialist in UML model bidirectional C code generation.
- Representative to the Object Management Group.

Software Engineer V — Northrop Grumman

September 2004 – September 2006 · Denver, CO

- Key software engineer on a Northrop Grumman Silver-award IR&D program.
- Network protocol stack programming, network security implementation and test, kernel programming, and FreeBSD system administration.
- Specialization in packet content filtering, redirection, and spoofing.
- XSLT, XPath, Java, JNI, C, C++, Cloudshield RAVE, and FreeBSD.

Independent Consulting — Real-Time, Embedded, and Model-Driven Systems Engineering

June 2000 – March 2007 · Colorado Front Range

Successive engagements at firms building real-time operating systems, synchronous reactive control systems, embedded conditional-access platforms, and UML-based software-engineering tooling:

- **Vidiom** (September 2006 – February 2007, Westminster, CO): Senior systems modeler and software engineer on the OpenCable Downloadable Conditional Access System — UML 2.0 design and C implementation of a messaging broker coupling Java-resident OCAP applications to 7816-resident SecureMicro Conditional Access clients.

- **Deadline Specialists** (February 2007 – December 2008, Boulder, CO, overlapping with the Artisan consulting engagement above): Contract embedded real-time OS programming on disk-drive firmware (Borland Together, Microsoft Visual Studio, CruiseControl, Ant; unit testing via `unit++`, CppUnit, mbUnit).
- **Design_Net Engineering** (December 2003 – August 2004, Denver, CO): UML modeling, UML-to-C generation, VxWorks AE on PowerPC, data acquisition and telemetry management, SQUID control.
- **Aztek Engineering** (July 2003 – November 2003, Denver, CO): UML modeling and C++ coding for GR-303 Local Digital Terminal.
- **Esterel Technologies** (May 2003 – July 2003, Denver, CO): Applications engineering and technical sales support for the Esterel SCADE synchronous design tool for dynamic and reactive control systems.
- **Artisan Software** (August 2002 – December 2002, Denver, CO): Applications engineering and technical sales support; UML modeling for real-time systems engineering (Artisan Studio, later Integrity Modeler).
- **QNX Software Systems** (April 2001 – February 2002, Denver, CO): QNX Neutrino applications engineering, technical sales support, IPv4 network stack programming.
- **TimeSys Real-time Linux** (June 2000 – April 2001, Denver, CO): Linux kernel programming, GNU GCC compiler builds, Linux distribution builder, technical sales support and marketing, Rate Monotonic Analysis consulting.

Member of Technical Staff — AT&T Bell Labs

January 1988 – December 1998 · Naperville, IL

- Unix and Linux kernel developer.
- Programmer of device drivers for network cards, flash memory, and Direct Memory Access controllers.
- Architect of a dual-node high-availability Linux racked server.
- Contributed to cellular technology platforms; recognized by peers for sustained technical excellence and for staying ahead of rapidly evolving standards across embedded and communications software domains.

Volunteer Service

Technical Program Director — INCOSE Colorado Front Range Chapter

September 2008 – September 2010 · Denver, CO

- Identified, recruited, and coordinated speakers of interest to practicing systems engineers for monthly chapter presentations and biannual tutorials.

Grand Knight — Knights of Columbus, Council 6905

August 2012 – July 2014

Deputy Grand Knight — Knights of Columbus, Council 6905

September 2009 – August 2010

- Served as the Grand Knight's right hand; chair of the membership retention committee; past Chancellor of the Council.

Education

Years	Degree	Institution
1986 – 1988	**M.S., Computer Engineering**	University of Illinois, Urbana-Champaign
1982 – 1986	**B.S., Electrical and Computer Engineering**	University of Illinois, Urbana-Champaign

Continuing Education

Year	Course	Provider
2025	Foundry & AIP Builder Foundations	Palantir Technologies
2025	Palantir Foundry Overview	Palantir Technologies
2024	Machine Learning Specialization	DeepLearning.AI / Stanford University (Coursera)
2022	Introduction to Cyber Security Learning Path	TryHackMe
2013	Functional Programming Principles in Scala	École Polytechnique Fédérale de Lausanne (Coursera)

Certifications, Clearances, and Standards Contributions

- U.S. Security Clearance (recent)
- OMG-Certified Business Process Modeler (OCBP)
- OMG BPMN2 Certified Modeler
- Neo4j Certified Professional (2020)
- Contributing Author — OMG UML Certification Exam
- Contributing Author — OMG BPMN Certification Exam
- Co-Chair — OMG Unified Profile for DoDAF and MODAF (UPDM)
- Co-Chair — INCOSE Model Lifecycle Management Working Group
- Technical Program Director — INCOSE Colorado Front Range Chapter (2008–2010)
- Trademark holder — DPYC™, Tollbooth DPYC™, Don't Pester Your Customer™

Technical Skills

Domain	Technologies
Architecture	Enterprise Architecture, MBSE, SysML v1 / v2, UML, BPMN, UPDM, UAF, DoDAF / MODAF, Digital Threads, OSLC, Model Context Protocol
Cybersecurity	CMMC, NIST 800-series, STIGs, FIPS, Risk Management Framework, CycloneDX SBOM, Snyk, OWASP Dependency-Check, OAuth2, SAML2, OIDC, Schnorr signatures, NIP-44 / NIP-17 encrypted messaging
Languages	Python, Swift / SwiftUI, Scala, Java, JavaScript / TypeScript, Groovy, Ruby, C, C++; HTTP / REST, GraphQL, SPARQL, Cypher, Gremlin
Data and Formats	JSON, XML, XMI, RDF / OWL, Turtle, Linked Data Platform, CSV, Neo4j, Cassandra, Neon (Postgres), pgvector
Blockchain and Identity	Bitcoin Lightning (BTCPay Server, LND, BOLT-11), Nostr (npubs, kind-30079, NIP-17, NIP-44, NIP-98), Schnorr signatures, OpenTimestamps
Platforms and Infrastructure	AWS (EC2, ECS Fargate, Lambda, Cognito, Amplify, Bedrock, API Gateway, CloudFormation, ECR, Marketplace), Prefect Horizon, Shopify, Gelato, Stripe, GitHub Actions, Docker, Kubernetes, Ruby Vagrant
MBSE and PLM Tooling	Syndeia, MagicDraw / Cameo Enterprise Architecture, Teamwork Cloud, TheBrain, Windchill PDMLink, IBM Jazz DNG / GCM, ParaMagic, DOORS, Artisan Studio / Integrity Modeler
Real-Time and Embedded	QNX Neutrino, VxWorks, Esterel SCADE, FreeBSD, embedded Linux, Rate Monotonic Analysis
Open Source	thebrain-mcp, schwab-mcp, excalibur-mcp, taxsort-mcp, optionality-mcp, tollbooth-sample, tollbooth-dpyc (Python wheel), tollbooth-authority, dpyc-oracle, tollbooth-oauth2-collector, tollbooth-shortlinks

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Languages

- English — Native
- French — Professional

Affiliations and Community

- Member, INCOSE (International Council on Systems Engineering)
- Active contributor to OMG working groups (SysML v2 API & Services, UPDM)
- Presenter at MBSE symposia, INCOSE chapter events, NDIA, and the Intercax Digital Thread Conference series
- Founder and curator of the DPYC Network State open-source community (`lonniev/dpyc-community` on GitHub)
- 500+ connections on LinkedIn; peer-recommended across telecommunications, defense, and systems engineering disciplines

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Lonnie VanZandt · 205 Ridgeline Road, Panton, VT 05491 · lonniev@gmail.com · April 2026